safeExam.Ai — System Architecture & Technical Report

Enterprise AI Proctoring & Examination System | Powered by Agent ELI (Custom Trained YOLOv11)

1. Executive Summary

safeExam.Ai is an automated online examination and real-time AI proctoring platform designed to ensure absolute academic integrity in remote testing environments. The system features a dual-shield defense architecture: **Agent ELI**, a custom-trained **YOLOv11 computer vision model** running directly on the candidate's browser at 1 FPS, coupled with deterministic client-side browser interaction guards. This eliminates video streaming bandwidth overhead while enforcing instant violation penalties, automated 10-demerit thresholds, and live teacher surveillance.

2. System Architecture & Technology Stack

Component Layer	Technology / Framework	Key Responsibilities
Frontend UI	React 18, Vite, Tailwind CSS	Single Page Application, camera stream, countdown modals
Computer Vision Core	Custom Trained YOLOv11 Model	On-device 1 FPS object detection (phone, book, presence)
Browser Event Guards	Web APIs (WebRTC, Beacon, Timing)	Traps shortcuts, reloads, context menu, and tab switches
Backend Server	Node.js v20+, Express.js REST API	JWT authentication, atomic demerits, 10s heartbeat monitor
Database Tier	MongoDB Atlas, Mongoose ODM	Persistent audit logs, student attempts, questions catalog

3. Agent ELI - Proctoring & Anti-Cheating Rules Matrix

#	Violation Trigger	Detection Logic	Penalty & Enforcement Action
1	Shortcut & Mouse Interaction	Ctrl+C/V/X/A, right-click, select-all	1st time: Warning Modal (0 pts); 2nd time+: +1 Demerit Point each time.
2	Browser Reload / Refresh	F5, Ctrl+R, browser refresh button	Immediate Auto-Submit via Beacon API. Locked out upon reload.
3	Tab Switch / Window Blur	visibilitychange, window blur event	+2 Demerits. If away >10s: 5s countdown modal -> Auto-Submit.
4	Network Disconnection	window.offline, 10s heartbeat loss	Exam on Hold (answering locked); auto-submits if offline >5 minutes.
5	Mobile Phone Detected	YOLOv11 phone detection >=5s	Warning modal with 5s countdown -> Auto-submits if not confirmed.
6	Book / Notebook Detected	YOLOv11 book detection	+1 Demerit point added for every 5 seconds visible.
7	Multiple Persons Detected	YOLOv11 person count > 1	+2 Demerit points added every 5 seconds.
8	Camera Absence / Black Screen	0 persons detected for >10s	Warning modal with 5s countdown -> Auto-submits exam.
9	10 Demerit Points Ceiling	Cumulative penalty accumulator	Immediate Auto-Submit upon reaching 10 demerit points.

4. Teacher Live Surveillance Interface

• **Right Column (Student Roster Tabs):** Interactive card tab buttons displaying live statuses (*In Exam*, *Submitted*, *Offline (Net Lost)*) when heartbeat >18s old, and demerit counters (X/10 pts). Includes master 'All Students Feed' button.

• **Left Column (AI Incident Feed):** Dynamically filters to the selected student card tab. Shows color-coded badges (Blue: Join, Emerald: Submitted, Red: Reload/Violation, Amber: Tab/Net, Purple: Shortcut).

5. System Verification & Build Health

• Production build verified via Vite (npm run build) completed with 0 errors.
• Backend operating on Port 5000 with real-time heartbeat monitoring and 5-minute auto-submit.
• Client bandwidth consumes <5 kbps per candidate.